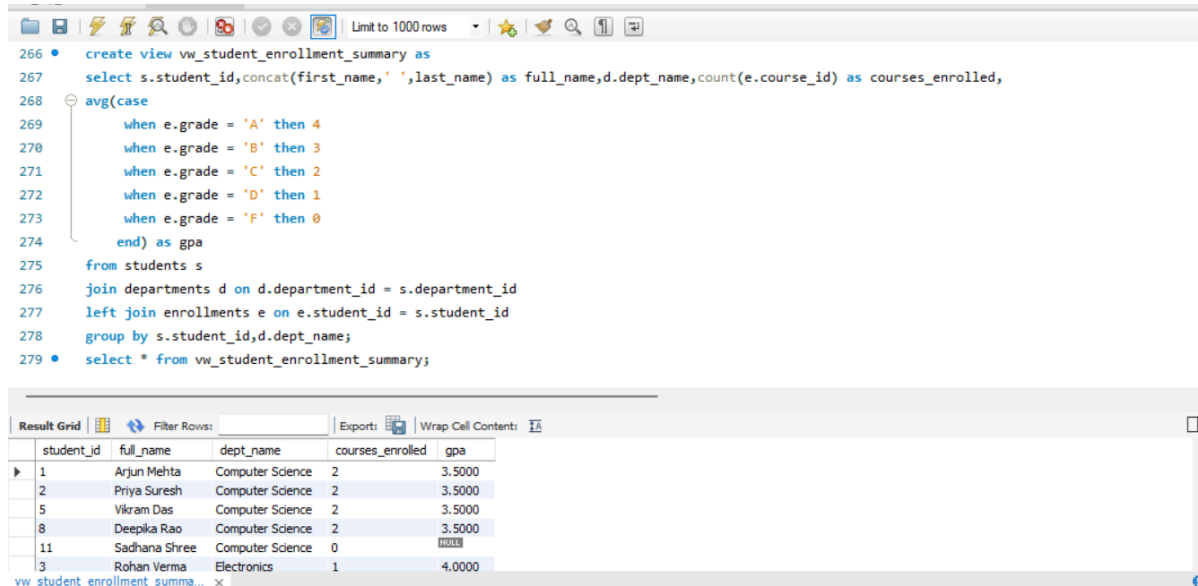


Task 2

Create a view vw_student_enrollment_summary showing each student's full name, department, number of courses enrolled in, and GPA (average grade converted: A=4, B=3, C=2, D=1, F=0).



```
266 • create view vw_student_enrollment_summary as
267 select s.student_id,concat(first_name,' ',last_name) as full_name,d.dept_name,count(e.course_id) as courses_enrolled,
268 avg(case
269     when e.grade = 'A' then 4
270     when e.grade = 'B' then 3
271     when e.grade = 'C' then 2
272     when e.grade = 'D' then 1
273     when e.grade = 'F' then 0
274 end) as gpa
275 from students s
276 join departments d on d.department_id = s.department_id
277 left join enrollments e on e.student_id = s.student_id
278 group by s.student_id,d.dept_name;
279 • select * from vw_student_enrollment_summary;
```

student_id	full_name	dept_name	courses_enrolled	gpa
1	Arjun Mehta	Computer Science	2	3.5000
2	Priya Suresh	Computer Science	2	3.5000
5	Vikram Das	Computer Science	2	3.5000
8	Deepika Rao	Computer Science	2	3.5000
11	Sadhana Shree	Computer Science	0	NULL
13	Rohan Verma	Electronics	1	4.0000

Create a view vw_course_stats showing course_name, course_code, total_enrollments, and avg_gpa for each course.

SQL File 5*

```

282 • create view vw_course_stats as
283   select course_name, course_code, count(e.enrollment_id) as total_enrollment,
284   avg(case
285     when e.grade = 'A' then 4
286     when e.grade = 'B' then 3
287     when e.grade = 'C' then 2
288     when e.grade = 'D' then 1
289     when e.grade = 'F' then 0
290   end
291   ) as avg_gpa
292   from courses c left join enrollments e on e.course_id = c.course_id
293   group by course_name, course_code;
294
295 • select * from vw_course_stats;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

course_name	course_code	total_enrollment	avg_gpa
Data Structures & Algorithms	CS101	4	3.5000
Database Management Systems	CS102	2	3.5000
Object Oriented Programming	CS103	2	3.5000
Circuit Theory	EC101	2	3.5000
Thermodynamics	ME101	0	0.0000

Query vw_student_enrollment_summary to find students with GPA above 3.0.

```

295
296   -- Query vw_student_enrollment_summary to find students with GPA above 3.0.
297 • select * from vw_student_enrollment_summary where gpa > 3.0;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

student_id	full_name	dept_name	courses_enrolled	gpa
1	Arjun Mehta	Computer Science	2	3.5000
2	Priya Suresh	Computer Science	2	3.5000
5	Vikram Das	Computer Science	2	3.5000
8	Deepika Rao	Computer Science	2	3.5000
3	Rohan Verma	Electronics	1	4.0000

Attempt to UPDATE a row through vw_student_enrollment_summary and note what happens.

Research and document in your comments why multi-table views are generally not updatable.

```
299 -- Attempt to UPDATE a row through vw_student_enrollment_summary and note what happens.
300 -- Research and document in your comments why multi-table views are generally not updatable.
301 • update vw_student_enrollment_summary
302 set gpa = 4
303 where student_id = 1;
```

Output

#	Time	Action	Message
28	12:03:26	select * from vw_courses_stats LIMIT 0, 1000	Error Code: 1146. Table 'college_db.vw_courses_stats' doesn't exist
29	12:03:38	select * from vw_course_stats LIMIT 0, 1000	5 row(s) returned
30	12:07:15	select * from vw_student_enrollment_summary where gpa > 3.0 LIMIT 0, 1000	5 row(s) returned
31	12:12:02	update vw_student_enrollment_summary set gpa = 4 where student_id = 1	Error Code: 1288. The target table vw_student_enrollment_summary of the UPDATE is not updatable

Attempting to update vw_student_enrollment_summary results in an error because the view is not updatable.

Reasons:

1. The view is based on multiple tables.
2. It uses aggregate functions (COUNT and AVG).
3. It uses GROUP BY.
4. A row in the view does not correspond to a single row in any one base table, so MySQL cannot determine which underlying table should be updated.

DROP both views and recreate vw_student_enrollment_summary as a view WITH CHECK OPTION

(use a single-table subset view for this step)

```

309 • CREATE VIEW vw_student_enrollment_summary AS
310 SELECT
311     student_id,
312     first_name,
313     last_name,
314     email,
315     enrollment_year
316 FROM students
317 WHERE enrollment_year >= 2023
318 WITH CHECK OPTION;
319 • SELECT * FROM vw_student_enrollment_summary;
320 • UPDATE vw_student_enrollment_summary
321 SET enrollment_year = 2023
322 WHERE id = 1;

```

Result Grid Filter Rows: Export: Wrap Cell Content:					
	student_id	first_name	last_name	email	enrollment_year
▶	4	Sneha	Patel	sneha.patel@college.edu	2023
	7	Aditya	Singh	aditya.singh@college.edu	2023
	11	Sadhana	Shree	sadhana.shree@college.edu	2023
	12	Neha	Gupta	neha.gupta@college.edu	2023

```

315     enrollment_year
316 FROM students
317 WHERE enrollment_year >= 2023
318 WITH CHECK OPTION;
319 • SELECT * FROM vw_student_enrollment_summary;
320 • UPDATE vw_student_enrollment_summary
321 SET enrollment_year = 2024
322 WHERE student_id = 4;
323

```

Result Grid Filter Rows: Export: Wrap Cell Content:					
	student_id	first_name	last_name	email	enrollment_year
▶	4	Sneha	Patel	sneha.patel@college.edu	2024
	7	Aditya	Singh	aditya.singh@college.edu	2023
	11	Sadhana	Shree	sadhana.shree@college.edu	2023
	12	Neha	Gupta	neha.gupta@college.edu	2023